(17) $\lim_{x\to 0}\frac{e^{\sin x}-1}{x}=$

(1) 1 (2) e (3) $\frac{1}{e}$ (4) 0

(18) $\lim_{x\to 0}\frac{e^{\tan x}-e^{x}}{\tan x-x}=$

(1) 1 (2) e (3) $\frac{1}{2}$ (4) 0

(19) The value of $\lim_{x\to 0}\frac{\sin x}{\sqrt{x^2}}$ is

(1) 1 (2) – 1 (3) 0 (4) limit does not exist

(20) The value of $\lim_{x\to k^-} x-\lfloor x\rfloor$, where k is an integer is

(1) – 1 (2) 1 (3) 0 (4) 2

(21) At $x=\frac{3}{2}$ the function $f(x)=\frac{|2x-3|}{2x-3}$ is

(1) continuous (2) discontinuous (3) differentiable (4) non-zero

(22) Let $f:\mathbb{R}\to\mathbb{R}$ be defined by $f(x)=\begin{cases}x & x \text{ is irrational}\\ 1-x & x \text{ is rational}\end{cases}$ then f is

(1) discontinuous at $x=\frac{1}{2}$ (2) continuous at $x=\frac{1}{2}$

(3) continuous everywhere (4) discontinuous everywhere

(23) The function $f(x)=\begin{cases}\frac{x^2-1}{x^3+1} & x\neq -1\\ P & x=-1\end{cases}$ is not defined for $x=-1$. The value of $f(-1)$ so that the function extended by this value is continuous is

(1) $\frac{2}{3}$ (2) $-\frac{2}{3}$ (3) 1 (4) 0

(24) Let f be a continuous function on [2, 5]. If f takes only rational values for all x and $f(3)=12$, then $f(4.5)$ is equal to

(1) $\frac{f(3)+f(4.5)}{7.5}$ (2) 12 (3) 17.5 (4) $\frac{f(4.5)-f(3)}{1.5}$

(25) Let a function f be defined by $f(x)=\frac{x-|x|}{x}$ for $x\neq 0$ and $f(0)=2$. Then f is

(1) continuous nowhere (2) continuous everywhere

(3) continuous for all x except $x = 1$ (4) continuous for all x except $x = 0$

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